

## Mark schemes

### Q1.

| Marking Instructions                                                                                               | AO     | Marks | Typical Solution                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------------------------------------------------------------------------------------------|
| Identifies and clearly defines consistent variables for length and width. Can be shown on diagram.                 | AO3.1b | B1    | Width of rectangle = $2x$<br>Length of rectangle = $2y$                                                    |
| Models the area of rectangle with an expression of the correct dimensions                                          | AO3.3  | M1    | $A = 4xy$                                                                                                  |
| Eliminates either variable to form a model for the area in one variable.                                           | AO1.1a | M1    | $x^2 + y^2 = 16$                                                                                           |
| Obtains a correct equation to model the area in one variable                                                       | AO1.1b | A1    | $A = 4x\sqrt{16 - x^2}$                                                                                    |
| Differentiates their expression for area. Condone one error                                                        | AO3.4  | M1    | $\frac{dA}{dx} = 4\sqrt{16 - x^2} - \frac{4x^2}{\sqrt{16 - x^2}}$                                          |
| Explains that their derivative equals zero for a maximum or stationary point.                                      | AO2.4  | E1    | $\frac{dA}{dx} = \frac{64 - 8x^2}{\sqrt{16 - x^2}}$<br>For maximum point $\frac{dA}{dx} = 0$               |
| Equates area derivative to zero and obtains correct value for either variable. CAO                                 | AO1.1b | A1    | $\frac{64 - 8x^2}{\sqrt{16 - x^2}} = 0$<br>$x = 2\sqrt{2}$                                                 |
| Completes a gradient test or uses second derivative of their area function to determine nature of stationary point | AO1.1a | M1    | When $x = 2.8$ , $\frac{dA}{dx} = 0.448$<br>When $x = 2.9$ , $\frac{dA}{dx} = -1.191$<br>Therefore maximum |
| Deduces that the area is a maximum at<br>$x = 2\sqrt{2}$ or $\theta = \frac{\pi}{4}$<br>Values need not be exact   | AO2.2a | R1    | The maximum area is 32 sq in                                                                               |
| Obtains maximum area with correct units AWRT 32                                                                    | AO3.2a | B1    |                                                                                                            |

**Total 10 marks**

**Q2.**

| Marking Instructions                                 | AO     | Marks | Typical Solution                                                                                                                                                     |
|------------------------------------------------------|--------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selects appropriate technique to differentiate       | AO3.1a | M1    | $2(x + y - 2)\left(1 + \frac{dy}{dx}\right) = e^y \frac{dy}{dx}$<br>$\frac{dy}{dx} = 0 \Rightarrow x + y - 2 = 0$<br>$\Rightarrow 0 = e^y - 1$<br>$y = 0$<br>$x = 2$ |
| Differentiates term involving $e^y$ correctly        | AO1.1b | B1    |                                                                                                                                                                      |
| Differentiates fully correctly                       | AO1.1b | A1    |                                                                                                                                                                      |
| Uses $\frac{dy}{dx} = 0$                             | AO1.1a | M1    |                                                                                                                                                                      |
| Eliminates $x$ or $y$ from the equation of the curve | AO1.1a | M1    |                                                                                                                                                                      |
| Obtains correct $y$ CAO                              | AO1.1b | A1    |                                                                                                                                                                      |
| Obtains correct $x$ CAO                              | AO1.1b | A1    |                                                                                                                                                                      |
|                                                      |        |       | <b>Total 7 marks</b>                                                                                                                                                 |

**Q3.**

|     | Marking Instructions                                                                                                                                                | AO     | Marks | Typical Solution                                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Deduces that the lower bound of $x$ is 1                                                                                                                            | AO2.2a | M1    | $\{x \in \mathbb{R} : x > 1\}$                                                                                                                                          |
|     | States the domain in a correct form                                                                                                                                 | AO2.5  | A1    |                                                                                                                                                                         |
| (b) | Differentiates using quotient rule (condone correct use of product rule)<br>Must have<br>$f'(x) = \frac{(2x-2)^{\frac{1}{2}} - kx(2x-2)^{-\frac{1}{2}}}{(2x-2)}$ OE | AO1.1a | M1    | $f'(x) = \frac{(2x-2)^{\frac{1}{2}} - \frac{1}{2}x(2x-2)^{-\frac{1}{2}} \times 2}{(2x-2)}$ $= \frac{2x-2-x}{(2x-2)^{\frac{3}{2}}}$ $= \frac{x-2}{(2x-2)^{\frac{3}{2}}}$ |
|     | Obtains correct derivative in unsimplified form                                                                                                                     | AO1.1b | A1    |                                                                                                                                                                         |
|     | Completes algebraic manipulation, with all previous working correct, to show the correct form. <b>AG</b>                                                            | AO2.1  | R1    |                                                                                                                                                                         |
|     |                                                                                                                                                                     |        |       |                                                                                                                                                                         |
| (c) | States that point of inflection requires second derivative to be 0                                                                                                  | AO2.4  | E1    | For point of inflection $f''(x) = 0$                                                                                                                                    |

|     |                                                                                                                                                |        |    |                                                                                                                                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | Forms an equation $f''(x) = 0$<br>OE                                                                                                           | AO1.1a | M1 | $f''(x) = \frac{(2x-2)^{\frac{3}{2}} - \frac{3}{2}(x-2)(2x-2)^{\frac{1}{2}} \times 2}{(2x-2)^3}$ $(2x-2)^{\frac{3}{2}} - 3(x-2)(2x-2)^{\frac{1}{2}} = 0$ $(2x-2)^{\frac{1}{2}}[(2x-2) - 3(x-2)] = 0$ $(2x-2)^{\frac{1}{2}}(4-x) = 0$ |
|     | Solves their equation                                                                                                                          | AO1.1a | M1 |                                                                                                                                                                                                                                      |
|     | Obtains solution $x = 4$                                                                                                                       | AO1.1b | A1 |                                                                                                                                                                                                                                      |
|     | Gives a valid reason for rejecting $x = 1$ , or cancels factor of $(2x-2)^{1/2}$ stating $x \neq 1$ .                                          | AO2.4  | E1 |                                                                                                                                                                                                                                      |
|     | Tests either side of 'their' $x = 4$                                                                                                           | AO1.1a | M1 |                                                                                                                                                                                                                                      |
|     | Completes rigorous argument to conclude they have one point of inflection<br><br>Do not award this mark if 2 <sup>nd</sup> E1 mark not awarded | AO2.1  | R1 | $x = 1$ or $x = 4$<br>$x \neq 1$ because of domain<br>$f''(3) = \frac{1}{32} > 0$<br>$f''(5) = \frac{-\sqrt{2}}{256} < 0$<br>Therefore point of inflection at $x = 4$                                                                |
| (d) | Deduces values of $x$ for convex section of graph                                                                                              | AO2.2a | B1 | $1 < x < 4$                                                                                                                                                                                                                          |
|     |                                                                                                                                                |        |    | <b>Total 13 marks</b>                                                                                                                                                                                                                |

**Q4.**

|      | Marking Instructions                                                                | AO     | Marks | Typical Solution                                                                                             |
|------|-------------------------------------------------------------------------------------|--------|-------|--------------------------------------------------------------------------------------------------------------|
| (a)  | Selects an appropriate routine procedure; evidence of quotient rule or product rule | AO1.1a | M1    | $\frac{dy}{dx} = \frac{2(4x^2 + 7) - 8x(2x + 3)}{(4x^2 + 7)^2}$                                              |
| (i)  | Obtains correct derivative (no need for simplification)                             | AO1.1b | A1    |                                                                                                              |
| (ii) | States clearly that $\frac{dy}{dx} > 0 \Rightarrow y$ is increasing                 | AO2.4  | R1    | $y$ is increasing $\Leftrightarrow \frac{dy}{dx} > 0$<br>$\frac{2(4x^2 + 7) - 8x(2x + 3)}{(4x^2 + 7)^2} > 0$ |
|      | Forms inequality from 'their' $\frac{dy}{dx} > 0$                                   | AO3.1a | B1    | $(4x^2 + 7)^2 > 0$ for all $x$<br>$\therefore 2(4x^2 + 7) - 8x(2x + 3) > 0$                                  |
|      | Deduces numerator must be positive                                                  | AO2.2a | R1    | $8x^2 + 14 - 16x^2 - 24x > 0$                                                                                |
|      | Considers denominator alone and sets out clear argument to justify given            | AO2.1  | R1    | $4x^2 + 12x - 7 < 0$ <b>(AG)</b>                                                                             |

|                      |                                                                                                                                                    |        |    |                                                                                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | inequality <b>AG</b><br>Only award this mark if they have a completely correct solution, which is clear, easy to follow and contains no slips      |        |    |                                                                                                                                                                           |
| (b)                  | Solves the correct quadratic inequality (accept evidence of factorising, completing the square, use of formula, or correct critical values stated) | AO1.1a | M1 | $(2x + 7)(2x - 1)$<br>$x = -\frac{7}{2}, \frac{1}{2}$                                                                                                                     |
|                      | Obtains fully correct answer, given as an inequality or using set notation                                                                         | AO1.1b | A1 | $-\frac{7}{2} < x < \frac{1}{2}$<br>$x \in \left(-\frac{7}{2}, \frac{1}{2}\right)$<br><b>Or</b><br>$x \in \left\{x : -\frac{7}{2} < x < \frac{1}{2}\right\}$<br><b>Or</b> |
| <b>Total 8 marks</b> |                                                                                                                                                    |        |    |                                                                                                                                                                           |

**Q5.**

|     | Marking Instructions                                                                                                                                                                                                                                                  | AO     | Marks | Typical Solution                                                                                                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Uses a correct method for finding $\frac{dy}{dx}$<br>evidence for this includes sight of $\frac{dy}{dx}$ or $\frac{dx}{dt}$ and chain rule<br>OR an attempt at implicit or explicit differentiation of a correct Cartesian equation or 'their' equation from part (b) | AO1.1a | M1    | $\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 2t$<br>$\frac{dy}{dx} = \frac{2t}{3t^2}$<br>When $t = -2$ $\frac{dy}{dx} = -\frac{1}{3}$<br><b>ALT</b><br>$y = (x - 2)^{\frac{2}{3}} - 1$<br>$\frac{dy}{dx} = \frac{2(x - 2)^{-\frac{1}{3}}}{3}$ |
|     | Obtains correct $\frac{dy}{dx}$                                                                                                                                                                                                                                       | AO1.1b | A1    | When $t = -2$ , $x = -6$<br>$\frac{dy}{dx} = \frac{2(-6 - 2)^{-\frac{1}{3}}}{3} = -\frac{1}{3}$                                                                                                                                          |
|     | Substitutes $t = -2$ (or $x = -6$ ) into 'their' equation for                                                                                                                                                                                                         | AO1.1a | M1    |                                                                                                                                                                                                                                          |

|                      |                                                                                                                                                                         |        |                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------|
|                      | $\frac{dy}{dx}$                                                                                                                                                         |        |                                                                                                 |
|                      | Obtains correct simplified gradient of the curve<br><br>FT 'their' equation for $\frac{dy}{dx}$                                                                         | AO1.1b | A1F                                                                                             |
| (b)                  | Eliminates $t$ or makes $t$ the subject in one expression (evidence for this includes one equation with $t$ as the subject or two equations with equal powers of $t$ .) | AO1.1a | M1                                                                                              |
|                      | Finds a correct Cartesian equation in any form                                                                                                                          | AO1.1b | A1                                                                                              |
|                      |                                                                                                                                                                         |        | <b>ALT</b><br>$t^3 = (x - 2)$<br>$t = (x - 2)^{\frac{1}{3}}$<br>$y = (x - 2)^{\frac{2}{3}} - 1$ |
| <b>Total 6 marks</b> |                                                                                                                                                                         |        |                                                                                                 |

**Q6.**

|     | Marking Instructions                                                        | AO     | Marks | Typical Solution                                                                                                                                                            |
|-----|-----------------------------------------------------------------------------|--------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Uses the product rule for either term                                       | AO1.1a | M1    | $y = 2x\cos 3x + (3x^2 - 4)\sin 3x$<br>$\frac{dy}{dx} = 2\cos 3x - 6x\sin 3x + 6x\sin 3x + 3(3x^2 - 4)\cos 3x$<br>$= (9x^2 - 10)\cos 3x$                                    |
|     | Uses the product rule for both terms                                        | AO1.1a | M1    |                                                                                                                                                                             |
|     | Differentiates both terms correctly                                         | AO1.1b | A1    |                                                                                                                                                                             |
|     | Rearranges to correct form CAO                                              | AO1.1b | A1    |                                                                                                                                                                             |
| (b) | Finds $\frac{d^2y}{dx^2}$ from 'their' first derivative and equates to zero | AO3.1a | M1    | $\frac{d^2y}{dx^2} = 18x\cos 3x - 3(9x^2 - 10)\sin 3x$<br><br>point of inflection $\Rightarrow \frac{d^2y}{dx^2} = 0 \Rightarrow$<br>$18x\cos 3x - 3(9x^2 - 10)\sin 3x = 0$ |
|     | Applies product rule correctly on 'their' $\frac{dy}{dx}$                   | AO1.1b | A1F   |                                                                                                                                                                             |

|                                                                                                                                                                                                              |        |     |                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|------------------------------------------------------------------|
| FT only applies if both M1 marks awarded in part (a)                                                                                                                                                         |        |     | $\Rightarrow \frac{\cos 3x}{\sin 3x} = \frac{3(9x^2 - 10)}{18x}$ |
| Arrives at a result using 'their' second derivative through correct algebraic manipulation that is correct for 'their' second derivative<br>FT only applies if both previous marks in (b) have been awarded. | AO1.1b | A1F | $\Rightarrow \cot 3x = \frac{9x^2 - 10}{6x} \quad (\mathbf{AG})$ |
| Constructs a clearly explained rigorous mathematical argument, to show the required result<br><br>This must include a concluding statement or an explanation of reasoning at the start. <b>AG</b>            | AO2.1  | R1  |                                                                  |
| <b>Total 8 marks</b>                                                                                                                                                                                         |        |     |                                                                  |

**Q7.**

|     | Marking Instructions                                             | AO     | Marks | Typical Solution                                                                                                                     |
|-----|------------------------------------------------------------------|--------|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Finds 2 <sup>nd</sup> derivative and sets up an inequality       | AO3.1a | M1    | $\frac{dy}{dx} = -2xe^{-x^2}$                                                                                                        |
|     | Obtains correct first derivative                                 | AO1.1b | A1    | $\frac{d^2y}{dx^2} = -2e^{-x^2} + 4x^2e^{-x^2}$                                                                                      |
|     | Obtains second derivative correct from 'their' first derivative  | AO1.1b | A1F   | $-2e^{-x^2} + 4x^2e^{-x^2} < 0$<br>$4x^2 - 2 < 0$<br>$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$                                  |
|     | Deduces correct final inequality (could use set notation)        | AO2.2a | A1    |                                                                                                                                      |
| (b) | Uses trapezium rule                                              | AO1.1a | M1    | $\int_{0.1}^{0.5} e^{-x^2} dx \approx \frac{0.1}{2} (e^{-0.01} + e^{-0.25} + 2(e^{-0.04} + e^{-0.09} + e^{-0.16}))$ $\approx 0.3611$ |
|     | Trapezium rule entries all correct                               | AO1.1b | A1    |                                                                                                                                      |
|     | Finds correct value                                              | AO1.1b | A1    |                                                                                                                                      |
| (c) | References area being completely within concave section<br>So... | AO2.4  | E1    | $[0.1, 0.5] \subset \left[ -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$<br>$\therefore$ area is completely within                |

|                       |                                                                                                                                                                                                       |        |    |                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----|-----------------------------------------------------------------------------------------------------------------------------|
|                       | Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)                                                                      | AO2.4  | E1 | concave section<br>Hence trapezia lie below curve and give an under-estimate for the area                                   |
| (d)                   | Uses suitable rectangle to obtain over-estimate                                                                                                                                                       | AO3.1a | B1 | Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate<br>Area of rectangle = |
|                       | Explains that this rectangle lies above the curve                                                                                                                                                     | AO2.4  | E1 |                                                                                                                             |
|                       | Constructs rigorous mathematical argument about accuracy, which leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips. | AO2.1  | R1 | $0.4 \times e^{-0.01^2} = 0.396\dots$<br>$\therefore 0.36 < A < 0.40$<br>So $A = 0.4$ to 1 dp                               |
| <b>Total 12 marks</b> |                                                                                                                                                                                                       |        |    |                                                                                                                             |

**Q8.**

(a)  $\left(\frac{dy}{dx}\right) = 3e^{3x} + \frac{1}{x}$

*B1 for one term correct*

B1

*B1 all correct*

B1

2

(b) (i)  $\left(\frac{du}{dx}\right) = \frac{\pm \cos x(1 + \cos x) \pm \sin x(\sin x)}{(1 + \cos x)^2}$

*clear attempt at quotient / product rule  
condone poor use of brackets*

M1

$$\frac{\cos x(1 + \cos x) - \sin x(-\sin x)}{(1 + \cos x)^2}$$

*any correct form seen*

A1

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \frac{1}{1 + \cos x}$$

AG be convinced  
 correct use of brackets and correct notation used  
 throughout (eg A0 if  $\cos x^2$  etc seen)

A1cso

3

(ii)  $\left(\frac{dy}{dx}\right) = \frac{1+\cos x}{\sin x} \times \frac{1}{1+\cos x}$  OE  
 correct use of chain rule

M1

$$= \frac{1}{\sin x}$$

$$= \operatorname{cosec} x$$

AG, must see  $= \frac{1}{\sin x}$  and no errors seen;  
 condone incorrect use of brackets only if penalised in part  
 (b)(i)

A1

2

[7]

**Q9.**

(a)  $y = 4x \cos 2x$

$$\left(\frac{dy}{dx}\right) = 4 \cos 2x - 4x(2) \sin 2x$$

anything reducible to  $A \cos 2x + Bx \sin 2x$  where  $A$  and  $B$   
 are non-zero integers

M1

gradient of the tangent

OE, all correct

A1

$$A \cos \frac{2\pi}{4} + B \times \frac{\pi}{4} \sin \frac{2\pi}{4}$$

substituting  $\frac{\pi}{4}$  into candidate's derived function

m1

$$= -2\pi$$



must have  $= -2\pi$  using correct  $\frac{dy}{dx}$

A1

an equation of the tangent is  $y = -2\pi \left( x - \frac{\pi}{4} \right)$

OE, dependent on previous A1

A1

5

(b)

$$\left( \int_0^{\frac{\pi}{4}} 4x \cos 2x \, dx \right)$$

$$u = Ax \quad \frac{dv}{dx} = \cos 2x$$

$$\frac{du}{dx} = A \quad v = B \sin 2x$$

all 4 terms in this form seen or used

M1

$A = 4$  and  $B = \frac{1}{2}$  or  $A = 1$  and  $B = 2$ , etc

A1

$$= \left[ 4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - \int_{(0)}^{\left(\frac{\pi}{4}\right)} 4 \times \frac{1}{2} \sin 2x \, (dx)$$

correct substitution of candidate's terms into integration  
by parts formula condone missing limits

m1

$$= \left[ 4x \frac{1}{2} \sin 2x \right]_{(0)}^{\left(\frac{\pi}{4}\right)} - [-\cos 2x]_{(0)}^{\left(\frac{\pi}{4}\right)}$$

candidate's second integration completed correctly  
FT on one error including coefficient condone missing limits

A1F

$$= \frac{\pi}{2} - 1$$

OE, exact value

A1

5

[10]

**Q10.**

(a)  $(y = x^4 \tan 2x)$

$$\left( \frac{dy}{dx} = \right) 4x^3 \tan 2x + x^4 2\sec^2 2x$$

$$4x^3 \tan 2x + Ax^4 \sec^2 kx$$

OE where  $A$  is a non-zero constant.

M1

A1 for  $k = 2$

may have  $(\sec 2x)^2$

$$\text{or } \frac{1}{\cos^2 2x}$$

A1

A1 all correct

ISW if attempt to simplify is incorrect.

A1

3

(b)  $\left( \frac{dy}{dx} = \right) \frac{\pm 2x(x-1) \pm 1(x^2)}{(x-1)^2}$

Use of the quotient rule

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M1

$$\frac{2x(x-1) - 1(x^2)}{(x-1)^2}$$

A1

$$\left( = \frac{x^2 - 2x}{(x-1)^2} \right)$$

**Simplification not required**

$$\left( \frac{dy}{dx} = \right) \frac{3}{4} \text{ or } 0.75 \text{ OE}$$

Obtained from correct  $\frac{dy}{dx}$

A1

Q11.

(a)  $\left(\frac{dx}{d\theta} = \right) \frac{(\sin \theta \times 0) - 1 \times \cos \theta}{\sin^2 \theta}$   
 quotient rule  $\frac{\pm \sin \theta \times k \pm 1 \times \cos \theta}{\sin^2 \theta}$  where  $k = 0$  or  $1$

M1

must see the '0' either in the quotient or in eg  $\frac{dx}{d\theta} = 0$  etc

A1

$$= -\frac{\cos \theta}{\sin^2 \theta} \quad \text{or} \quad = -\frac{\cos \theta}{\sin \theta \sin \theta}$$

or equivalent

$$= -\operatorname{cosec} \theta \cot \theta$$

CSO, AG must see one of the previous expressions

A1

(b)  $x = \operatorname{cosec} \theta$

$$\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$$

OE, eg  $dx = -\operatorname{cosec} \theta \cot \theta d\theta$

B1

Replacing  $\sqrt{(\operatorname{cosec}^2 \theta - 1)}$  by  $\sqrt{\cot^2 \theta}$ , or better  
 at any stage of solution

B1

$$\int = \int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \sqrt{(\operatorname{cosec}^2 \theta - 1)}} d\theta$$

all in terms of  $\theta$ , and including their attempt at  $dx$ ,  
 but condone omission of  $d\theta$

M1

fully correct and must include  $d\theta$  (at some stage in solution)

A1

$$\int \frac{-\operatorname{cosec} \theta \cot \theta}{\operatorname{cosec}^2 \theta \cot \theta} (d\theta) = \int \frac{-1}{\operatorname{cosec} \theta} (d\theta)$$

OE eg  $\int -\sin\theta (d\theta)$

A1

$= \cos\theta$

A1

$x = 2, \theta = 0.524$  AWRT

$x = \sqrt{2}, \theta = 0.785$  AWRT

correct change of limits or  $(\pm)\cos\theta = (\pm)\left[\sqrt{1-\frac{1}{x^2}}\right]_{\sqrt{2}}^2$  OE

B1

$0.8660 - 0.7071$

c's  $F(0.52) - F(0.79)$

substitution into  $\pm \cos\theta$  only or  $\left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right)$

m1

$= 0.159$

A1

9

[12]

**Q12.**

(a)  $\left(\frac{dy}{dx} =\right) x^3 \times \frac{1}{x} + 3x^2 \ln x$

$px^3 \times \frac{1}{x} + qx^2 \ln x$

where  $p$  and  $q$  are integers

M1

$p = 1, q = 3$

A1

2

(b) (i)  $\left(\frac{dy}{dx} =\right) e^2 + 3e^2 \ln e (= 4e^2)$

Substituting  $e$  for  $x$  in their  $\frac{dy}{dx}$ , but must have scored M1 in (a)

M1

$$y = e^3 \ln e (= e^3)$$

B1

$$y - e^3 = 4e^2 (x - e)$$

*OE but must have evaluated  $\ln e$  (twice) for this mark  
(must be in exact form, but condone numerical evaluation  
after correct equation)*

A1

3

$$(ii) -e^3 = 4e^2 (x - e) \text{ or } 4e^2 x = 3e^3 \quad OE$$

*Correctly substituting  $y = 0$  into a correct tangent  
equation in (b)(i)*

M1

$$x = \frac{3}{4} e$$

*CSO;*

*ignore subsequent decimal evaluation*

A1

2

[7]

**Q13.**

$$(a) \left( \frac{dx}{dy} \right) = \frac{\cos y \times \cos y - \sin y \times -\sin y}{\cos^2 y}$$

*Condone incorrect signs, poor notation, omission of  
 $\frac{dy}{dx}$  or using  $\frac{dx}{dy}$*

M1

$$= \frac{\cos^2 y + \sin^2 y}{\cos^2 y}$$

*RHS correct with terms squared, including correct notation  
Must see this line*

A1

$$= \frac{1}{\cos^2 y} \text{ or } (= 1 + \tan^2 y)$$

*Must see one of these*

$$\frac{dx}{dy} = \sec^2 y$$

AG; all correct including correct use of  $\frac{dx}{dy}$  throughout

A1  
CSO

3

(b)  $\sec^2 y = 1 + (x - 1)^2$

Correct use of  $\sec^2 y = 1 + \tan^2 y$  and in terms of  $x$

M1

$$= 1 + x^2 - 2x + 1 \quad \text{OE}$$

$$= x^2 - 2x + 2$$

AG; must see " $\sec^2 y =$ ",  $(x - 1)^2$  expanded and no errors seen

A1

2

(c)  $\frac{dx}{dy} = x^2 + 2x + 2$  or  $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

Must be seen

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

AG and no errors seen

B1

1

(d) (i)  $y = \tan^{-1}(x - 1) - \ln x$

$$\left( \frac{dy}{dx} = \right) \frac{1}{x^2 - 2x + 2} - \frac{1}{x}$$

Must be correct

M1

$$\left( \frac{dy}{dx} = 0 \right)$$

$$\pm x^2 + bx + c (= 0)$$

Expression in this form (generous), where  $b$  and  $c \neq 0$

m1

$$x^2 - 3x + 2 = 0$$

Must see correct equation = 0

A1

$$x = 1, 2$$

*Both answers must be seen*

*The two A marks are independent*

A1

4

(ii)  $y'' = p(x^2 - 2x + 2)^{-2}(2x - 2) \pm qx^{-2}$   
*where  $p$  and  $q$  are constants*

M1

$$y'' = -(x^2 - 2x + 2)^{-2}(2x - 2) + x^{-2}$$

$$p = -1, q = 1 \text{ including correct brackets}$$

A1

2

(iii)  $x = 1, y'' = 1$

*Must have scored full marks in (d)(i) and (ii)*

M1

At  $x = 1, y'' > 0 \therefore \text{min}$

When  $x = 1, y = 0$  hence on  $x$ -axis

*Must see  $y'' > 0$  or in words*

*Both statements fully correct*

A1

2

EXAM PAPERS PRACTICE

[14]

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**Q14.**

(a)  $\frac{dy}{dx} = k(x^3 - 1)^5$

*Where  $k$  is an integer or function of  $x$*

M1

$$= 6 \times 3x^2(x^3 - 1)^5$$

(ISW)

*But note*

$$\frac{dy}{dx} = k(x^3 - 1)^5 + px^2$$

M0

**Or**

$$(u = x^3 - 1) \quad (y = u^6)$$

$$\frac{dy}{du} = 6u^5 \text{ and } \frac{du}{dx} = 3x^2$$

M1

$$= 6(x^3 - 1)^5 \times 3x^2 \quad \text{A1}$$

*Note*

$$\frac{dy}{dx} = 6 \times 3x^2 (x^3 - 1)^5 + c$$

scores M1 A0

*(penalise + c in differential once only in paper)*

A1

2

(b) (i)  $\frac{dy}{dx} = \pm x \times \frac{1}{x} \pm \ln x$

*Product rule attempted and differential of  $\ln x$*

M1

$$= 1 + \ln x \quad \text{(ISW)}$$

A1

2

(ii)  $(x = e) \quad y = e \quad \text{PI}$

*Must have replaced  $\ln e$  by 1*

*Condone  $y = 2.72$  (AWRT)*

B1

$$\frac{dy}{dx} = 1 + \ln e (= 2)$$

$$\frac{dy}{dx}$$

*Correct substitution into their  $\frac{dy}{dx}$   
But must have scored M1 in (b)(i)*

M1

$$y - e = 2(x - e) \text{ or } y = 2x - e \quad \text{OE, ISW}$$

*Must have replaced  $\ln e$  by 1*

A1

3

[7]

**Q15.**

(a) (i)  $\frac{dx}{dy} = k \sec^2(3y + 1)$

*Where  $k$  is an integer*



Condone omission of  $\frac{dx}{dy}$

But

$$\frac{dy}{dx} = k \sec^2(3y+1) \quad \text{scores}$$

M1 A0

M1

$$= 3 \sec^2(3y+1) \quad \text{ISW}$$

A1

**Alternative methods**

$$y = \frac{1}{3}(\tan^{-1}x - 1)$$

$$\frac{dx}{dy} = k(1+x^2)$$

M1

$$3 = (1 + \tan^2(3y+1))$$

A1

**Or**

$$x = \frac{\sin(3y+1)}{\cos(3y+1)}$$

$$\frac{dx}{dy} = \frac{\pm k \cos^2(3y+1) \pm k \sin^2(3y+1)}{\cos^2(3y+1)}$$

M1

$$= \frac{3}{\cos^2(3y+1)}$$

A1

2

(ii)  $\frac{dx}{dy} = 3 \sec^2\left(3x - \frac{1}{3} + 1\right) = 3 \sec^2 0$

Substitution of  $y = -\frac{1}{3}$  into their  $\frac{dx}{dy}$  or  $\frac{dy}{dx}$

BUT must have scored M1 in (a)(i)

M1

$$\frac{dy}{dx} = \frac{1}{3} \quad \text{CSO}$$

Condone 0.333 or better

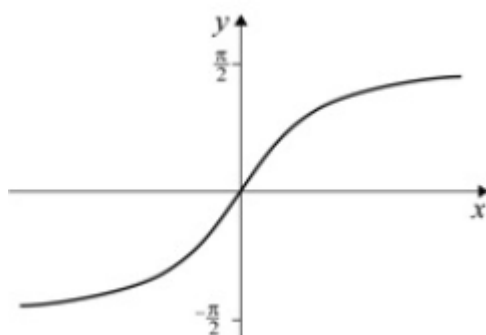
**Or**

$$\left. \begin{aligned} \frac{dy}{dx} &= \frac{1}{3\sec^2(3y+1)} \\ &= \frac{1}{3\sec^2 0} \\ &= \frac{1}{3} \end{aligned} \right\} \text{As above}$$

A1

2

(b)



*Approx correct shape with no turning points, through (0,0) and only 1 curve*

*Asymptotic at both  $\pm \frac{\pi}{2}$  and both values shown*

M1

*Condone  $\pm 90$  (degrees)*

*Condone  $y = \tan x$  also drawn but clearly identified, otherwise M0*

A1

2

[6]

**Q16.**

(a) (i)  $y = xe^{2x}$

$$\left( \frac{dy}{dx} \right) = 2xe^{2x} + e^{2x}$$

$kxe^{2x} + le^{2x}$  where  $k$  and  $l$  are 1s or 2s

M1

A1

$$\left. \begin{matrix} k = 2 \\ l = 1 \end{matrix} \right\} \text{Independent of each other}$$

A1

$$(= e^{2x} (2x + 1))$$

ISW

3

(ii)  $x = 1 \Rightarrow \frac{dy}{dx} = 3e^2$

correct substitution of  $x = 1$  into their  $\frac{dy}{dx}$  but must have earned M1 in part (i)

M1

tangent:  $y - e^2 = 3e^2(x - 1)$  OE

CSO (no ISW), must have scored first 4 marks

common correct answer:  $y = 3e^2 x - 2e^2$

A1

2

(b)  $y = \frac{2 \sin 3x}{1 + \cos 3x}$

$$\left( \frac{dy}{dx} = \right) \frac{(1 + \cos 3x)6 \cos 3x - 2 \sin 3x(-3 \sin 3x)}{(1 + \cos 3x)^2}$$

$$\pm \frac{p(1 + \cos 3x) \cos 3x \pm q \sin 3x (\sin 3x)}{(1 + \cos 3x)^2}$$

where  $p$  and  $q$  are rational numbers

condone poor use/omission of brackets

PI by further working

M1

$$= \frac{6 \cos 3x + 6 \cos^2 3x + 6 \sin^2 3x}{(1 + \cos 3x)^2}$$

this line must be seen in this form (ie in terms of  $\cos^2 3x$  and  $\sin^2 3x$ ), but allow  $\sin^2 3x$  replaced by  $1 - \cos^2 3x$

condone denominator correctly expanded

A1

correct use of  $k \sin^2 3x + k \cos^2 3x = k$

$$\text{or } k \sin^2 3x = k (1 - \cos^2 3x)$$

m1

$$= \frac{6 \cos 3x + 6}{(1 + \cos 3x)^2}$$

$$= \frac{6}{1 + \cos 3x}$$

CSO

A1

4

note: if degrees used then no marks in (a) and (c)

[9]

**Q17.**

(a)  $(C =) \frac{2}{e} \text{ or } 2e^{-1} \text{ or } 2\left(\frac{1}{e}\right) \text{ or } 2(e^{-1})$

*One of these answers only.  
Not 0.736 but allow ISW.*

B1

1

(b)  $\frac{d}{dx}(2y) = 2 \frac{dy}{dx}$

B1

$$\frac{d}{dx}(e^{2x} y^2) = 2e^{2x} y^2 + e^{2x} 2y \frac{dy}{dx}$$

*Product; 2 terms added, one with  $\frac{dy}{dx}$  ;*

M1

*A1 for each term*

A1

A1

$$\frac{d}{dx}(x^2 + C) = 2x$$

B1

$$\frac{dy}{dx} =$$

Solve their equation correctly for  $\frac{dy}{dx}$

M1

$$\frac{x - e^{2x}y^2}{e^{2x}y + 1}$$

Condone factor of 2 in both numerator and denominator. ISW

A1

7

(c) Evaluate  $\frac{dy}{dx}$  at  $\left(1, \frac{1}{e}\right)$

Substitute  $x = 1$  and  $y = \frac{1}{e}$  into numerator of  $\frac{dy}{dx}$  ;  
 allow one slip

M1

$$\text{numerator} = 1 - e^2 e^{-2} = 0 \Rightarrow \text{stationary point}$$

Conclusion required; must score full marks in part (b)

Allow  $1 - 1 = 0$  or  $2 - 2 = 0$

A1

2

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[10]

**Q18.**

(a)  $y' = e^{-4x}(2x + 2) - 4e^{-4x}(x^2 + 2x - 2)$

$$y' = Ae^{-4x}(ax + b) \pm Be^{-4x}(x^2 + 2x - 2)$$

where  $A$  and  $B$  are non-zero constants

M1

All correct

A1

$$= e^{-4x}(2x + 2 - 4x^2 - 8x + 8)$$

$$\text{or } -4x^2e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

AG; all correct with no errors,  
 2<sup>nd</sup> line (OE) must be seen  
 Condone incorrect order on final line

A1

or

$$y = x^2 e^{-4x} + 2xe^{-4x} - 2e^{-4x}$$

$$y' = -4x^2 e^{-4x} + 2xe^{-4x} + 2x \cdot -4e^{-4x} + 2e^{-4x} + 8e^{-4x}$$

$$Ax^2 e^{-4x} + Bxe^{-4x} + Cxe^{-4x} + De^{-4x} + Ee^{-4x}$$

(M1)

All correct

(A1)

$$= -4x^2 e^{-4x} - 6xe^{-4x} + 10e^{-4x}$$

$$= 2e^{-4x}(5 - 3x - 2x^2)$$

AG; all correct with no errors,  
 3<sup>rd</sup> line (OE) must be seen

(A1)

3

(b)  $-(2x + 5)(x - 1) (= 0)$

OE Attempt at factorisation  
 $(\pm 2x \pm 5)(\pm x \pm 1)$   
 or formula with at most one error

M1

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$$x = -\frac{5}{2}, 1$$

Both correct and no errors  
 SC  $x = 1$  only scores M1A0

A1

$$x = 1, y = e^{-4}$$

For  $y = ae^b$  attempted

m1

Either correct, follow through only from  
 incorrect sign for  $x$

A1F

$$x = -\frac{5}{2}, y = e^{10} \left( -\frac{3}{4} \right)$$

CSO 2 solutions only

Note: withhold final mark for extra solutions

Note: approximate values only for y can score m1 only

A1

5

[8]

**Q19.**

(a) 
$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

M1

$$= \frac{4 \cos^2 4x + 4 \sin^2 4x}{\cos^2 4x} \text{ or better}$$

Both terms correct

A1

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

A1

or

$$\left(\frac{dy}{dx}\right) = \frac{\cos 4x \cdot 4 \cos 4x - \sin 4x \cdot -4 \sin 4x}{\cos^2 4x}$$

$$\frac{\pm A \cos^2 4x \pm B \sin^2 4x}{\cos^2 4x}$$

(M1)

$$= \frac{4 \cos 4x \cos 4x}{\cos 4x \cos 4x} + \frac{4 \sin 4x \sin 4x}{\cos 4x \cos 4x} \text{ or better}$$

(A1)

$$= 4(1 + \tan^2 4x)$$

CSO

All correct

(A1)

(b)  $\frac{d^2 y}{dx^2} = 4 \times 2 \tan 4x \times \dots$   
 $A \tan 4x \times f(4x)$

M1

$4 \sec^2 4x$   
 $f(4x) = B \sec^2 4x$

m1

$= 32 \tan 4x \sec^2 4x$   
*ft 8 × their p from part (a)*

A1F

$= 32 \tan 4x (1 + \tan^2 4x)$   
*Previous two method marks must have been earned*

m1

$= 32y(1 + y^2)$   
 CSO

A1

#### Alternative Solutions

$y' = 4 + 4 \tan^2 4x = 4 + 4 \frac{\sin^2 4x}{\cos^2 4x}$   
 $y'' = 4 \times \left[ \frac{\cos^2 4x \cdot 2 \sin 4x \cdot 4 \cos 4x + \sin^2 4x \cdot 2 \cos 4x \cdot 4 \sin 4x}{\cos^4 4x} \right]$

$\frac{A \cos^3 4x \pm B \sin^3 4x}{\cos^4 4x}$  where  $A$  and  $B$  are constants or trig functions.

(M1)

Where  $A$  is  $m \sin 4x$  and  $B$  is  $n \cos 4x$

(m1)

$= \frac{4 \times 8 \sin 4x \cos 4x [\cos^2 4x + \sin^2 4x]}{\cos^4 4x}$   
*ft 8 × their p from part (a)*



(A1F)

$$= 32 \tan 4x \sec^2 4x$$

$$k \tan 4x \sec^2 4x$$

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4 \sec^2 4x$$

$$\frac{d^2 y}{dx^2} = 4 \times 2 \sec 4x \cdot 4 \sec 4x \tan 4x$$

$$A \sec 4x \times f(4x)$$

(M1)

$$f(4x) = B \sec 4x \tan 4x$$

(m1)

$$= 32 \sec^2 4x \tan 4x$$

ft 8 × their p from part (a)

(A1F)

$$= 32(1 + \tan^2 4x) \tan 4x$$

Previous two method marks must have been earned

(m1)

$$= 32y(1 + y^2)$$

CSO

(A1)

or

$$\frac{dy}{dx} = 4(1 + \tan^2 4x)$$

$$u = \tan 4x \quad \frac{dy}{dx} = 4 + 4u^2$$

$$\frac{d^2 y}{dx^2} = (8)u \frac{du}{dx}$$

(M1)

$$\frac{du}{dx} = 4 + 4 \tan^2 4x = 4 + 4u^2$$

(m1)

$$\frac{d^2y}{dx^2} = 8u(4 + 4u^2)$$

(A1)

$$= 32u(1 + u^2)$$

(m1)

$$= 32y(1 + y^2)$$

(A1)

5

[8]

**Q20.**

$$x^2 + xy = e^y$$

$$2x + y + x \frac{dy}{dx} = e^y \frac{dy}{dx}$$

$2x$

B1

*Use product rule*

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M1A1

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B1

$$(-1, 0) \quad \frac{dy}{dx} = -1$$

CSO

A1

[5]

**Q21.**

(a) (i)  $\sin 2\theta = 2 \sin \theta \cos \theta$

B1

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

*OE condone use of  $x$  etc, but variable must be consistent*

B1

2

$$(ii) \quad \sin \theta = \frac{4}{5} \Rightarrow \sin 2\theta = 2 \times \frac{4}{5} \times \frac{3}{5} = \frac{24}{25}$$

AG

*Use of  $106.26^\circ$  .... B0*

B1

or

$$2 \times \sin\left(\cos^{-1} \frac{3}{5}\right) \times \frac{3}{5}$$

$$\cos 2\theta = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$$

$-0.28$

B1

2

$$(b) \quad (i) \quad \frac{dx}{d\theta} = 6 \cos 2\theta, \quad \frac{dy}{d\theta} = -8 \sin 2\theta$$

*Attempt both derivatives. ie  $p \cos 2\theta$*

M1

*Both correct.*

*$q \sin 2\theta$*

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A1

$$\frac{dy}{dx} = -\frac{4 \sin 2\theta}{3 \cos 2\theta} \quad \text{ISW}$$

CSO OE

A1

3

$$(ii) \quad P\left(\frac{72}{25}, -\frac{28}{25}\right)$$

$(2.88, -1.12)$

B1F

$$\text{Gradient} = -\frac{4}{3} \times -\frac{24}{7}$$

$$\text{Their } \frac{q \sin 2\theta}{p \cos 2\theta} \quad \text{or} \quad \frac{p \cos 2\theta}{q \sin 2\theta}$$

*must be working with rational numbers*

M1

Tangent  $y + \frac{28}{25} = \frac{32}{7} \left( x - \frac{72}{25} \right)$  ISW

*Any correct form.*

$7y = 32x - 100$

*Fractions in simplest form*

*Equation required*

A1

3

[10]

**Q22.**

(a) (i)  $y = \ln(5x - 2)$

$$\frac{k}{5x - 2}$$

M1

$$\left( \frac{dy}{dx} \right) = \frac{5}{5x - 2}$$

No ISW, eg  $\frac{5}{5x - 2} = \frac{1}{x - 2}$  (M1A0)

A1

2

(ii)  $y = \sin 2x$

$$\left( \frac{dy}{dx} \right) = 2 \cos 2x$$

A1

2

$k \cos 2x$

M1

(b) (i)  $f(x) \geq \ln 0.5$  or  $f(x) \geq -\ln 2$

M1A1

2

- (ii)  $(gf(x) =) \sin [2 \ln (5x - 2)]$   
 or  $(gf(x) =) \sin \ln (5x - 2)^2$   
*Condone*  
 *$\sin 2 \ln (5x - 2)$  or  $\sin 2 (\ln (5x - 2))$*   
*but **not**  $\sin 2 (\ln 5x - 2)$  or  $\sin 2 \ln 5x - 2$*

B1

1

- (iii)  $gf(x) = 0$   
 $\sin[2 \ln(5x - 2)] = 0$   
 $2 \ln(5x - 2) = 0$   
*Correct first step from their (b)(ii)*

M1

$$5x - 2 = 1$$

*Their  $f(x) = 1$  from  $k \ln (f(x)) = 0$*

m1

$$x = \frac{3}{5}$$

*Withhold if clear error seen other than omission of brackets*

A1

3

- (iv)  $x = \sin 2y$   
 $\sin^{-1} x = 2y$  (or  $\sin^{-1} y = 2x$ )  
*Correct equation involving  $\sin^{-1}$*

M1

$$(g^{-1}(x) =) \frac{1}{2} \sin^{-1} x$$

A1

2

[12]

**Q23.**

(a)  $y = \frac{\ln x}{x}$

(when)  $y = 0 \quad x = 1 \quad \text{or} \quad (1, 0)$

*Both coordinates must be stated, not 1 simply shown on diagram*

B1

1

(b)  $\left(\frac{dy}{dx}\right) = \frac{x \times \frac{1}{x} - \ln x}{x^2}$

Quotient/product rule  $\frac{\pm \frac{x}{x} \pm \ln x}{x^2}$

M1

$$= \frac{1 - \ln x}{x^2} \text{ or } x^{-2} - x^{-2} \ln x$$

OE must simplify  $\frac{x}{x}$

A1

At B,  $\frac{1 - \ln x}{x^2} = 0$

Putting their  $\frac{dy}{dx} = 0$  or numerator = 0

m1

$x = e$

CSO condone  $x = e^1$

A1

$y = \frac{1}{e} \text{ or } e^{-1}$

CSO must simplify  $\ln e$

A1

5

(c) Gradient at  $x = e^3$

$$= \frac{1 - \ln e^3}{(e^3)^2}$$

Substituting  $x = e^3$  into their  $\frac{dy}{dx}$  (condone 1 slip) but must have scored M1 in (b)

M1

$$= \frac{-2}{e^6} \text{ or } -2e^{-6}$$

*PI*

A1

Gradient of normal  $= \frac{1}{2} e^6$   
*CSO simplified to this*

A1

3

[9]

**Q24.**

(a)  $y = e^{2x}(x^2 - 4x - 2)$

$$\frac{dy}{dx} = e^{2x}(2x - 4)$$

*Product rule; allow 1 slip*

M1

$$+ (x^2 - 4x - 2)2e^{2x}$$

A1

$$\frac{dy}{dx} = e^{2x}(2x - 4 + 2x^2 - 8x - 4)$$

*Factorising  $e^{2x}(2x^2 - 6x - 8)$*

M1

$$e^{2x}(2x^2 - 6x - 8)$$

$$\text{Or } x^2 - 3x - 4 = 0$$

A1

$$e^{2x} \neq 0$$

$$(x - 4)(x + 1) = 0$$

*Solving 3 term quadratic  
 Dependent on both M marks*

m1

$$x = 4, -1$$

*And no extras eg  $x = 0$*

A1

(b) (i)  $\frac{d^2y}{dx^2} = e^{2x} \cdot 2 + (2x - 4)2e^{2x}$

$\frac{dy}{dx}$   
Product rule from their  $\frac{dy}{dx}$  in form  
 $e^{2x}$  (quadratic)  $e^{2x}(4x^2 - 8x - 22)$

M1

$$+ (x^2 - 4x - 2)4e^{2x} + 2e^{2x}(2x - 4)$$

A1

**OR**

$$\frac{d^2y}{dx^2} = e^{2x}(4x - 6) + (2x^2 - 6x - 8)2e^{2x}$$

M1A1

2

(ii)  $x = 4 : y'' = e^8 (10) > 0 \therefore \text{MIN}$

$\frac{d^2y}{dx^2}$   
Their 2 x's in their  $\frac{d^2y}{dx^2}$   
only of form  $e^{2x}$  (quadratic)

M1

$x = -1 : y'' = e^{-2} (-10) < 0 \therefore \text{MAX}$

CSO Both correct  
Allow values either side of y or y'

A1

2

[10]

**Q25.**

(a)  $P(-1, \pi)$

Condone  $(-1, 180^\circ)$

B1

$Q(1, 0)$

B1

2

(b) Translate



E1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

or equivalent in words

B1

Stretch SF 2 // y-axis

Stretch + one other correct

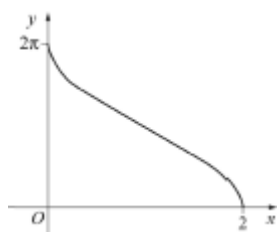
M1

all correct

A1

4

(c)



Correct shape in 1st quadrant

$2\pi$  and 2 marked correctly

B1

B1

2

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(d) (i)

$$\frac{y}{2} = \cos^{-1}(x - 1)$$

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M1

$$\cos\left(\frac{y}{2}\right) = x - 1$$

$$x = \cos\left(\frac{y}{2}\right) + 1$$

A1

2

(ii)  $-\frac{1}{2} \sin\left(\frac{y}{2}\right)$

$k \sin(\dots)$

M1

$$\frac{dx}{dy} \text{ correct}$$

A1

$$\text{At } y = 2, \left( \frac{dx}{dy} = \right) - \frac{1}{2} \sin 1$$

Condone AWR T -0.42

A1

3

[13]

**Q26.**

$$(a) \quad y = \frac{4x}{4x-3}$$

$$\frac{dy}{dx} = \frac{(4x-3) \cdot 4 - 4x(4)}{(4x-3)^2}$$

*Must use quotient rule*  
*Condone one slip*

M1

$$= \frac{-12}{(4x-3)^2}$$

$$k = -12$$

A1

2

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$$(b) \quad (i) \quad y = x \ln(4x-3)$$

$$\frac{dy}{dx} = \frac{x \cdot 4}{4x-3} + \ln(4x-3)$$

$$\frac{f(x)}{4x-3} + g(x) \quad 'f(x)' \text{ may be constant}$$

M1

$$\frac{kx}{4x-3} + \ln(4x-3)$$

m1A1

3

$$(ii) \quad x = 1 \quad y = 0$$

B1

$$\frac{dy}{dx} = 4$$

Sub  $x = 1$  into their  $\frac{dy}{dx}$

M1

$\therefore y = 4(x - 1)$       any correct form  
 CSO Must have full marks in (b)(i)

A1

3

(c) (i)  $u = 4x - 3$   
 $du = 4 dx$

M1

$$\int \frac{4x}{4x-3} dx = \int \frac{u+3}{u} \frac{du}{4}$$

Or  $\int \frac{4x}{4x-3} dx = \int \left(1 + \frac{3}{4x-3}\right) dx$

A1

$$= \left(\frac{1}{4}\right) \int \left(1 + \frac{3}{u}\right) (du)$$

$$= \int \left(1 + \frac{3}{u}\right) du \text{ etc}$$

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m1

$$= \frac{1}{4} (u + 3 \ln u)$$

$$= \frac{1}{4} [(4x - 3) + 3 \ln (4x - 3)] (+c)$$

CSO Condone missing  $du$

A1

4

(ii)  $\int \ln(4x - 3) dx$

$$u = \ln(4x - 3) \quad \frac{dv}{dx} = 1$$

*In correct direction*

M1

$$\frac{du}{dx} = \frac{4}{4x-3} \quad v = x$$

$$\int x \ln(4x-3) - \int \frac{4x}{4x-3} dx$$

A1

$$= x \ln(4x-3) - \frac{1}{4} [(4x-3) + 3 \ln(4x-3)]$$

*x \ln(4x-3) - their (c)(i)*

m1A1

(+ c)

4

[16]

**Q27.**

(a) (i)  $f(x) = \frac{\cos x}{2x+1} - \frac{1}{2}$   
*OE*

$$f(0) = \frac{1}{2}; f\left(\frac{\pi}{2}\right) = -\frac{1}{2}$$

$$x=0 \quad LHS=1, \quad x=\frac{\pi}{2} \quad LHS=0$$

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Change of sign  $0 < \alpha < \frac{\pi}{2}$

*Either side of  $\frac{1}{2}, \therefore 0 < \alpha < \frac{\pi}{2}$*

A1

2

(ii)  $\frac{\cos x}{2x+1} = \frac{1}{2}$

$$\left. \begin{array}{l} 2\cos x = 2x+1 \\ 2\cos x - 1 = 2x \end{array} \right\} \text{or, } \cos x = x + \frac{1}{2}$$

*Either line*

$$x = \cos x - \frac{1}{2}$$

$\cos x - \frac{1}{2} = x$   
 AG; or  
 All correct with no errors

B1

1

(iii)  $x_1 = 0$

$x_2 = 0.5$

Attempt at iteration

(allow  $x_2 = -0.5, x_3 = 0.38, 0.4$ )

M1

$x_3 = 0.378$

CAO

A1

2

(b) (i)  $\frac{dy}{dx} = \frac{(2x+1)(-\sin x) - \cos x \times 2}{(2x+1)^2}$

Attempt at quotient rule:

$$\frac{\pm (2x+1)\sin x \pm 2\cos x}{(2x+1)^2}$$

M1

Either term correct

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A1

All correct ISW

A1

3

(ii)  $x = 0$

$$\frac{dy}{dx} = -2$$

Correctly subst.  $x = 0$  into their  $\frac{dy}{dx}$

m1

$$\therefore \text{Gradient of normal} = \frac{1}{2}$$

CSO

A1

2

[10]

**Q28.**

$$8x + 2y \frac{dy}{dx} = 3y + 3x \frac{dy}{dx}$$

8x and 4  $\rightarrow 0$

B1

$$2y \frac{dy}{dx}$$

B1

$$3y + 3x \frac{dy}{dx}$$

Two terms with one  $\frac{dy}{dx}$

M1A1

at (1, 3) (gradient)  $\frac{dy}{dx} = \frac{1}{3}$

CSO

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[5]

**Q29.**

(a) (i)  $y = (2x^2 - 5x + 1)^{20}$

$$\frac{dy}{dx} = 20(2x^2 - 5x + 1)^{19}(4x - 5) \quad \text{OE}$$

chain rule 20 ( )<sup>19</sup> f(x)

M1

with no further incorrect working

A1

2

(ii)  $y = x \cos x$

$$\frac{dy}{dx} = -x \sin x + \cos x$$

*product rule  $\pm x \sin x \pm \cos x$*

M1

CSO

A1

2

(b)  $y = \frac{x^3}{x-2}$

$$\frac{dy}{dx} = \frac{(x-2)3x^2 - x^3 \times 1}{(x-2)^2}$$

*quotient rule  $\frac{\pm uv' \pm u'v}{(x-2)^2}$*

M1

*condone missing brackets*

A1

$$= \frac{3x^2 - 6x^2 - x^3}{(x-2)^2}$$

$$= \frac{2x^2(x-3)}{(x-2)^2}$$

CSO

A1

3

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[7]

**Q30.**

$$3x \frac{dy}{dx} + 3y - 4y \frac{dy}{dx} = 0$$

*attempt implicit differentiation*

M1

*product*

A1

*chain*

A1

*constant*

B1

$$\frac{dy}{dx} = -\frac{3}{2}$$

CSO

A1

**ALTERNATIVE METHOD**

$$x = \frac{2}{3}y + \frac{4}{3y}$$

*solve for  $x =$  expression in  $y$  and  
differentiate with respect to  $y$*

(M1)

$$\frac{dx}{dy} = \frac{2}{3} - \frac{4}{3y^2}$$

(A1A1)

$$y = 1, \frac{dx}{dy} = \frac{2}{3} - \frac{4}{3}$$

*substitute  $y = 1$*

(M1)

$$\frac{dx}{dy} = -\frac{3}{2}$$

CSO

(A1)

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[5]

**Q31.**

$$(a) \quad \frac{dy}{dx} = 5(3x + 1)^4 \times 3$$

$$k(3x + 1)^4$$

M1

$$= 15(3x + 1)^4$$

*with no further errors (w.n.f.e)*

A1



(b)  $\frac{dy}{dx} = \frac{3}{3x+1}$

$$\frac{k}{3x+1}$$

*w.n.f.e*

M1

A1

2

(c)  $\frac{dy}{dx} = (3x+1)^5 \times \frac{3}{3x+1} + \ln(3x+1) \times 15(3x+1)^4$

*product rule  $uv' + u'v$  (from (a) and (b))*

M1

*either term correct*

A1

*CSO with no further errors*

A1

$$\left( \begin{array}{l} = (3x+1)^4 [3 + 15 \ln(3x+1)] \\ = 3(3x+1)^4 [1 + 5 \ln(3x+1)] \end{array} \right)$$

3

[7]

**Q32.**

(a)  $\frac{dy}{dx} = -x^2 \sin 2x + \cos 2x$

*product rule  $kx \sin 2x \pm \cos 2x$*

M1

*no further incorrect working*

A1

2

(b) (i)  $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

*replacing  $x = \alpha$  and writing equation equal to zero (at any line)*

M1

$$\left. \begin{aligned} 2\alpha \sin 2\alpha &= \cos 2\alpha \\ 2\alpha \tan 2\alpha &= 1 \end{aligned} \right\} \text{either}$$

$$2\alpha \tan 2\alpha - 1 = 0$$

AG; CSO

A1

2

(ii)  $\left. \begin{aligned} f(0.4) &= 0.2 \\ f(0.5) &= -0.6 \end{aligned} \right\} \text{awrt} \quad \text{o.e}$

(0.9's unsubstantiated scores M0)

M1

Change of sign  $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii)  $2x \tan 2x = 1$

$$\left. \begin{aligned} \tan 2x &= \frac{1}{2x} \\ 2x &= \tan^{-1}\left(\frac{1}{2x}\right) \end{aligned} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv)  $x_1 = 0.4$

$$x_2 = 0.4480\dots$$

$$x_2 = 25.7$$

M1

$$\begin{aligned} x_3 &= 0.4200\dots \\ &= 0.42 \end{aligned}$$

A1

2

(c)  $y = x \cos 2x$

$$\left. \begin{aligned} u &= x & du &= 1 \\ dv &= \cos 2x & v &= \frac{\sin 2x}{2} \end{aligned} \right\}$$

$\left. \begin{array}{l} \text{differentiate one term} \\ \text{integrate one term} \end{array} \right\} \text{ must be } k \sin 2x$

M1

correct substitution of their values into  
 parts formula using  $u = x$

m1

$$\begin{aligned}
 \int &= \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx) \\
 &= \left[ \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)}
 \end{aligned}$$

A1

$$= \left( \frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left( \frac{\cos 0}{4} \right)$$

correctly substituting values from  
 previous 2 method marks

m1

$$= 0.0954$$

AWRT

A1

5

[14]

Q33.

EXAM PAPERS PRACTICE

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(a)  $y = \frac{\sin \theta}{\cos \theta}$

$$\begin{aligned}
 \frac{dy}{d\theta} &= \frac{\cos \theta \cos \theta - \sin \theta (-\sin \theta)}{\cos^2 \theta} \\
 &= \frac{\pm \cos^2 \theta \pm \sin^2 \theta}{\cos^2 \theta}
 \end{aligned}$$

M1A1

$$= \frac{1}{\cos^2 \theta} \quad \text{o.e.}$$

$$(1 + \tan^2 \theta)$$

$$= \sec^2 \theta$$

AG; CSO

A1

**Alternative:**

$$y = \frac{\tan \theta}{1}$$

$$\frac{dy}{d\theta} = \frac{1 \sec^2 \theta - 0}{1^2}$$

M1A1

$$= \sec^2 \theta$$

A1

3

(b)

$$x = \sin \theta$$

$$x^2 = \sin^2 \theta$$

$$\cos^2 \theta = 1 - x^2$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$= \frac{x}{\sqrt{1 - x^2}}$$

**OR LHS =**

$$\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta$$

use of  $\cos^2 \theta + x^2 = 1$  M1

AG; CSO

A1

2

(c)

$$\int \frac{1}{(1 - x^2)^{\frac{3}{2}}} dx$$

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

o.e.

$$\frac{dx}{d\theta} = \pm \cos \theta$$

M1

$$\int = \int \frac{\cos \theta (d\theta)}{(1 - \sin^2 \theta)^{\frac{3}{2}}}$$

all in terms of  $\theta$

m1

$$= \int \frac{\cos \theta}{(\cos^2 \theta)^{\frac{3}{2}}} (d\theta)$$

A1

$$= \int \sec^2 \theta (d\theta)$$

A1

$$= \tan \theta$$

$$= \frac{x}{\sqrt{1-x^2}} (+c)$$

CSO including  $d\theta$ 's

A1

5

[10]

**Q34.**

(a) (i)  $y = (4x^2 + 3x + 2)^{10}$

$$\frac{dy}{dx} = 10(4x^2 + 3x + 2)^9 (8x + 3)$$

For  $f(x)^9$  where  $f(x) \neq k$  and is linear

M1

A1

2

(ii)  $y = x^2 \tan x$

Product rule

M1

$$\frac{dy}{dx} = x^2 \sec^2 x + 2x \tan x$$

A1

2

(b) (i)  $x = 2y^3 + \ln y$

$$\frac{dx}{dy} = 6y^2 + \frac{1}{y}$$

B1

1

(ii) At (2,1)

$$\frac{dx}{dy} = 6 + 1 = 7$$

M1

$$\frac{dy}{dx} = \frac{1}{7}$$

*May be implied*

A1 ✓

$$(y-1) = \frac{1}{7}(x-2)$$

OE

A1

3

[8]

**Q35.**

(a) (i)  $\int \frac{dy}{y} = \int \sin t \, dt$

*Attempt to separate and integrate*

M1

$$\ln y = -\cos t + c$$

*A1 for  $\ln y$ ; A1 for  $-\cos t$ ; condone missing  $c$*

A1, A1

$$y = ae^{-\cos t}$$

*A present; or  $y = e^{-\cos t + C}$*

A1

4

(ii)  $y = 50, t = \pi: 50 = Ae^{-\cos \pi} = Ae$

*Substitute  $y = 50, t = \pi$  to find constant*

*Can have  $50 = e^{1+C}$  if substituted in above*

$$e^C = \frac{50}{e}$$

M1

A1

$$y = 50e^{-1}e^{-\cos t}$$

*AG (convincingly obtained)*

A1

3

**Alternative:**

Must have a constant in answer to (a)(i)

$$y = Ae^{-\cos t} \text{ or } y = e^{-\cos t + c} \text{ or } \ln y = -\cos t + c$$

$$50 = Ae^{-\cos \pi} \quad 50 = e^{-\cos \pi + c} \quad \ln 50 = -\cos \pi + c$$

(M1)

$$50 = Ae^{-1} \quad 50 = e^{-1+c} \quad \ln y = -\cos t + \ln 50 - 1$$

(A1)

$$y = 50e^{-1-\cos t} \quad y = e^{-\cos t} \quad \frac{50}{e} \ln\left(\frac{y}{50}\right) = -1 - \cos t$$

(A1)

**Alternative:**

Substitute  $y = 50$ ,  $t = \pi$  into  $\ln y = -\cos t + c$  M1

$$\ln y = -\cos t + \ln 50 - 1 \quad \text{A1}$$

$$\ln \frac{y}{50} = -1 - \cos t \quad \text{(AG)} \quad \text{A1}$$

(b) (i)  $t = 6: y = 50e^{-1-\cos 6} = 7.0417... \approx 7\text{cm}$

Degrees 6.8 SC1

7 or 7.0 for A1

M1A1

2

(ii)  $t = \pi \Rightarrow (\sin t = 0 \Rightarrow) \frac{dy}{dt} = 0$

Condone  $x$  for  $t$

B1

$$\frac{d^2 y}{dt^2} = y \cos t + \frac{dy}{dt} \sin t$$

For attempt at product rule including  $\frac{dy}{dt}$

term; must have  $\frac{d^2 y}{dt^2} =$

M1

$$t = \pi$$

A1

$$\frac{d^2 y}{dt^2} = y \cos \pi + \frac{dy}{dt} \sin \pi$$

$$= -50 \Rightarrow \max$$

*Accept = -y, with explanation that y is never negative*

A1

4

**Alternative:**

$$y = 50 e^{-(1+\cos t)} = \frac{50}{e} e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e} e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2 y}{dt^2} = \frac{50}{e} e^{-\cos t} \times \cos t + \frac{50}{e} e^{-\cos t} \times \sin^2 t$$

*Attempt at product rule*

(M1)

*Correct*

(A1)

$$\text{Substitute } t = \pi \rightarrow -50 \Rightarrow \max$$

(A1)

[13]